Chi-Square Tests & Regression Inference

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Chi-square goodness-of-fit test**

A die is rolled 60 times, expecting each face to appear equally often (expected count 10 per face). The observed counts are $12, 8, 11, 9, 7, 13$. Compute the chi-square test statistic using $\chi^2 = \sum \frac{(O-E)^2}{E}$.

$\chi^2 = \frac{(12-10)^2}{10} + \frac{(8-10)^2}{10} + \frac{(11-10)^2}{10} + \frac{(9-10)^2}{10} + \frac{(7-10)^2}{10} + \frac{(13-10)^2}{10} = \frac{4+4+1+1+9+9}{10} = \frac{28}{10} = 2.8.$

2 **[Foundational]** • **Target: Chi-square goodness-of-fit test**

State the null and alternative hypotheses for the goodness-of-fit test in the die-rolling scenario above.

H_0: the die is fair (each face has probability $\frac{1}{6}$). H_a: the die is not fair (at least one face's true probability differs from $\frac{1}{6}$).

3 **[Foundational]** • **Target: Chi-square test for independence**

A two-way table examines whether preferred learning style (visual, auditory, kinesthetic) is associated with academic major (STEM, non-STEM). State the null and alternative hypotheses for a chi-square test of independence in this context.

H_0: preferred learning style and academic major are independent (no association). H_a: preferred learning style and academic major are not independent (there is an association).

4 **[Foundational]** • **Target: Chi-square test for independence**

In a chi-square test of independence with a 3×2 table (3 learning styles, 2 major categories), find the degrees of freedom, using $df = (\text{rows} - 1)(\text{columns} - 1)$.

$df = (3-1)(2-1) = 2 \times 1 = 2.$

5 **[Foundational]** • **Target: Interpreting chi-square results**

A chi-square test of independence produces a test statistic $\chi^2 = 8.2$ with $df = 2$, and a p-value of 0.017. State a conclusion at the $\alpha = 0.05$ significance level, in context.

Since the p-value (0.017) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence of an association between the two categorical variables being studied.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Interpreting chi-square results

Explain a condition that must be checked before conducting a chi-square test, related to expected cell counts.

All expected counts in the table should be at least 5 (a common guideline). If some expected counts are too small, the chi-square approximation may not be reliable, and results should be interpreted with caution or a different method used.

7 [Foundational] • Target: Inference for the slope of a regression line

A regression of exam score on hours studied gives a sample slope $b_1 = 6.2$ with standard error $SE_{b_1} = 1.1$, based on $n = 25$ data points. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.069$ (with $df = 23$).

Margin of error: $2.069(1.1) \approx 2.28$. Confidence interval: 6.2 ± 2.28, or approximately $(3.92, 8.48)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inference for the slope of a regression line

State the null and alternative hypotheses for testing whether hours studied has a significant linear relationship with exam score, and compute the test statistic using the values from the previous problem.

$H_0 : \beta_1 = 0$ (no linear relationship). $H_a : \beta_1 \neq 0$ (there is a linear relationship). Test statistic: $t = \frac{b_1 - 0}{SE_{b_1}} = \frac{6.2}{1.1} \approx 5.64$.

9 [Intermediate] • Target: Conditions for regression inference

State the four conditions (sometimes remembered as LINER: Linear, Independent, Normal, Equal variance, Random) required for valid inference about a regression slope.

(1) Linear: the true relationship between x and y is linear. (2) Independent: individual observations are independent of each other. (3) Normal: for any fixed x-value, the responses y vary according to a normal distribution. (4) Equal variance: the standard deviation of y is the same for all values of x (no fanning pattern in the residuals). Additionally, data should come from a properly randomized design or random sample.

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Putting it together: a full inference scenario**

A study finds a chi-square test statistic of $\chi^2 = 3.1$ with $df = 1$, and a critical value of $\chi^2_{0.05} = 3.84$ at the $\alpha = 0.05$ level. State the conclusion, and explain how this differs from a p-value based approach.

Since the test statistic (3.1) is less than the critical value (3.84), we fail to reject H_0 — there is not enough evidence of an association. Using critical values compares the test statistic directly to a threshold on the chi-square distribution, while a p-value approach compares a computed probability to α directly; both lead to the same reject/fail-to-reject decision when done correctly, since the critical value is simply the test-statistic value corresponding to a p-value of exactly α.

11 **[Intermediate]** • **Target: Chi-square test for homogeneity**

Explain the difference between a chi-square test of independence and a chi-square test of homogeneity, even though both use the same formula and table structure.

A test of independence uses a single sample and asks whether two categorical variables measured on the same individuals are associated. A test of homogeneity compares the distribution of a single categorical variable across two or more separate populations or groups (e.g. comparing the distribution of preferred learning style between two different schools) — the calculation and table setup are identical, but the sampling design and the question being asked differ.

12 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

In a two-way table, a row total is 80, a column total is 120, and the grand total is 300. Find the expected count for that cell, using $E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$.

$E = \frac{(80)(120)}{300} = \frac{9600}{300} = 32.$

13 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

State the degrees of freedom for a chi-square goodness-of-fit test with 6 categories, using $df = (\text{number of categories}) - 1$.

$df = 6 - 1 = 5.$

14 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

A genetics experiment expects offspring to appear in a 9:3:3:1 ratio across four phenotype categories, based on 160 total offspring observed. Find the expected count for each category, and briefly describe what the chi-square goodness-of-fit test would examine using these expected counts.

Total ratio parts: $9 + 3 + 3 + 1 = 16$. Expected counts: $\frac{9}{16}(160) = 90$, $\frac{3}{16}(160) = 30$, $\frac{3}{16}(160) = 30$, $\frac{1}{16}(160) = 10$. The chi-square goodness-of-fit test would compare these expected counts to the actual observed counts in each category, computing $\chi^2 = \sum \frac{(O-E)^2}{E}$ to assess whether the observed offspring distribution is consistent with the predicted 9:3:3:1 genetic ratio.

© ClipSAT — clipsat.org | All Rights Reserved

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Computing expected counts for a chi-square test** • **Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A study examines the relationship between hours of sleep per night (x) and reaction time in milliseconds (y) for 30 participants, finding a regression slope $b_1 = -8.5$ with standard error $SE_{b_1} = 2.3$. State the hypotheses for testing whether more sleep is associated with a significant decrease in reaction time (a one-sided test), compute the test statistic, and interpret the sign of the slope in context.

$H_0 : \beta_1 = 0$ (no linear relationship). $H_a : \beta_1 < 0$ (more sleep is associated with decreasing reaction time). Test statistic: $t = \frac{b_1 - 0}{SE_{b_1}} = \frac{-8.5}{2.3} \approx -3.70$. The negative slope indicates that, in this sample, reaction time tends to decrease (improve) as hours of sleep increase — consistent with the idea that more sleep is associated with faster reaction times.

16 **[Challenging]** • **Target: Computing expected counts for a chi-square test** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A university surveys 240 students, recording both their preferred study location (library, dorm, coffee shop) and their class year (underclassman, upperclassman), to test whether study location preference is independent of class year. The resulting chi-square test statistic is $\chi^2 = 11.4$ with $df = 2$, and the corresponding p-value is 0.0033. State the conclusion at $\alpha = 0.01$, and explain, in the context of this scenario, what rejecting or failing to reject H_0 would mean for the university's understanding of its students.

Since the p-value (0.0033) is less than $\alpha = 0.01$, we reject H_0. This means there is convincing statistical evidence of an association between preferred study location and class year — in context, this suggests underclassmen and upperclassmen tend to have systematically different study location preferences, which could inform decisions like where the university invests in study space improvements for each group.

© ClipSAT — clipsat.org | All Rights Reserved

17 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A researcher studies the relationship between advertising spending (in thousands of dollars, x) and monthly sales (in thousands of dollars, y) for 22 small businesses, obtaining a regression slope $b_1 = 3.2$ with standard error $SE_{b_1} = 0.85$. Construct a 95% confidence interval for the true population slope β_1 using a critical value $t^* \approx 2.086$ (with $df = 20$), and interpret the interval in context, including whether it suggests advertising spending has a statistically significant positive association with sales.

Margin of error: $2.086(0.85) \approx 1.773$. Confidence interval: 3.2 ± 1.773, or approximately $(1.427, 4.973)$. Interpretation: we are 95% confident that, for every additional thousand dollars spent on advertising, monthly sales increase by somewhere between about 1.427 and 4.973 thousand dollars, on average. Since this entire interval is positive (it does not contain 0), this suggests a statistically significant positive association between advertising spending and sales.

18 [Challenging] • Target: Chi-square test for independence • Cross-Lesson: 3.03 The Normal Distribution

Level 2 · MediumA significance test produces a test statistic of $z = 2.15$ for a two-sided test $H_0 : p = 0.40$ vs $H_a : peq 0.40$. What is the p-value, and what decision is made at $\alpha = 0.05$?

For a standard normal distribution, $P(Z > 2.15) = 1 - 0.9842 = 0.0158$. Since the alternative hypothesis is two-sided (eq), multiply by 2: $\text{p} - \text{value} = 2 \times 0.0158 = \mathbf{0.0316}$. Because x. There is convincing statistical evidence that the true population proportion differs from 0.40.

19 [Challenging] • Target: Interpreting chi-square results • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Eight patients with chronic insomnia try a new relaxation therapy. Their sleep duration (hours/night) is measured before and after 4 weeks of therapy: Patient differences $d = \text{After} - \text{Before}$ yield $\bar{x}_d = 1.25$ hours and $s_d = 0.80$ hours. Assuming normality of differences, test whether the therapy significantly increased sleep at $\alpha = 0.01$.

1. Method Choice: Paired data measured on the same 8 patients $\Rightarrow$ Paired t-test on differences. 2. Hypotheses: $H_0 : \mu_d = 0 \quad \text{vs.} \quad H_a : \mu_d > 0 \quad (\alpha = 0.01)$, where μ_d is the true mean difference in sleep duration (After minus Before). 3. Test Statistic & p-value: $t = \frac{\bar{x}_d - 0}{s_d/\sqrt{n}} = \frac{1.25 - 0}{0.80/\sqrt{8}} = \frac{1.25}{0.2828} \approx 4.42 \quad (df = 8 - 1 = 7)$. With $df = 7$, $P(T > 4.42) \approx 0.0015$. 4. Conclusion: Because.

20 [Challenging] • Target: Inference for the slope of a regression line • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

[Unit 4 · Statistical Inference: Proportions, Means & Slopes · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Chi-square goodness-of-fit test

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $17, 21, 13, 9$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 5.33 ✓

b. 4.33

c. 5.00

d. 6.33

$\chi^2 = (4 + 36 + 4 + 36)/15 = 80/15 \approx 5.33$.

22 [Foundational] • Target: Chi-square test for independence

In a two-way table, a row total is 87, a column total is 46, and the grand total is 200. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 23.01

b. 112.99

c. 20.01 ✓

d. 17.01

$E = (87)(46)/200 = 4002/200 \approx 20.01$.

23 [Foundational] • Target: Interpreting chi-square results

A regression gives a sample slope $b_1 = -3.5$ with standard error $SE_{b_1} = 1.1$, based on data with $df = 10$. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.228$.

a. $(-5.45, -1.05)$

b. $(-5.95, -1.05)$ ✓

c. $(-6.95, -0.05)$

d. $(-5.95, -1.55)$

Margin of error: $2.228(1.1) \approx 2.45$. Interval: -3.5 ± 2.45, or approximately $(-5.95, -1.05)$.

24 [Foundational] • Target: Inference for the slope of a regression line

A regression gives a sample slope $b_1 = 3.8$ with standard error $SE_{b_1} = 0.9$. Compute the test statistic for testing $H_0 : \beta_1 = 0$, using $t = b_1/SE_{b_1}$, to two decimal places.

a. 3.22

b. 4.22 ✓

c. 5.22

d. -4.22

$t = b_1/SE_{b_1} = 3.8/0.9 \approx 4.22$.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Conditions for regression inference

A chi-square test produces a test statistic $\chi^2 = 15.9$ with $df = 4$, and a critical value of $\chi^2_{0.05} = 9.49$ at $\alpha = 0.05$. State the conclusion, comparing the test statistic directly to the critical value.

a. The test cannot be completed without a p-value

b. **Reject H_0; there is convincing evidence of an association ✓**

c. Fail to reject H_0; there is not convincing evidence of an association

d. Accept H_0 as definitely true

Since the test statistic (15.9) is greater than the critical value (9.49), we reject H_0. Reject H_0; there is convincing evidence of an association.

26 [Foundational] • Target: Putting it together: a full inference scenario

A chi-square goodness-of-fit test has 4 categories with observed counts 21, 25, 50, 10 (equal expected proportions). Find the chi-square test statistic.

a. **32.34 ✓**

b. 35.34

c. 30.34

d. 36.34

$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(21-26.5)^2}{26.5} + \frac{(25-26.5)^2}{26.5} + \frac{(50-26.5)^2}{26.5} + \frac{(10-26.5)^2}{26.5} = 32.34.$

27 [Foundational] • Target: Chi-square test for homogeneity

A study investigates the effectiveness of a new teaching method by comparing the test scores of students who used the new method versus those who used the traditional method. The results are summarized in a two-way table. If the table has 4 rows and 3 columns, what is the degrees of freedom (df) for a chi-square test of homogeneity?

a. 12

b. **6 ✓**

c. 5

d. 2

For a chi-square test of homogeneity or independence with a two-way table, the degrees of freedom (df) are calculated using the formula: $\text{df} = (r-1)(c-1)$, where r is the number of rows and c is the number of columns. In this case, the number of rows is $r = 4$ and the number of columns is $c = 3$. Substituting these values into the formula: $\text{df} = (4-1)(3-1) = (3)(2) = 6$. Therefore, the degrees of freedom are 6.

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Computing expected counts for a chi-square test

A chi-square goodness-of-fit test has 4 categories. Find the degrees of freedom.

a. 4 **b.** 2

c. 5 **d.** **3** ✓

For goodness-of-fit, $df = k - 1 = 4 - 1 = 3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Chi-square goodness-of-fit test

A chi-square test of independence uses a 2×4 table. Find the degrees of freedom.

a. 4 **b.** 5

c. 2 **d.** **3** ✓

For a test of independence, $df = (r - 1)(c - 1) = (2 - 1)(4 - 1) = 3$.

30 [Intermediate] • Target: Chi-square test for independence

In a chi-square test of independence for a contingency table with 7 rows and 2 columns (Trial #5), what is the value of df?

a. **6** ✓ **b.** 7

c. 14 **d.** 13

$\text{df} = (r - 1)(c - 1) = (7 - 1)(2 - 1) = 6$.

31 [Intermediate] • Target: Interpreting chi-square results

In a test of independence with total sample $N = 390$, row total 85, and column total 70 (Experiment 5), find the expected cell count.

a. 14.38 **b.** 16.33

c. **15.26** ✓ **d.** 44.29

Expected frequency: $E = \frac{\text{Row Total} \times \text{Col Total}}{\text{Grand Total}} = \frac{85 \times 70}{390} = 15.26$.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Inference for the slope of a regression line

Chi-square test is used for:

a. Comparing two means

b. **Relationships between categorical variables ✓**

c. Regression coefficients

d. Differences in variances only

Chi-square tests for association between categoricals.

33 [Intermediate] • Target: Conditions for regression inference

Conditions for inference with proportions (1-prop z-test):

a. n>30

b. **np≥10 and n(1-p)≥10 ✓**

c. Normal population

d. Known σ

Large counts condition: np≥10 and n(1-p)≥10.

34 [Intermediate] • Target: Putting it together: a full inference scenario

A test of $H_0 : \mu = 81$ uses a sample of $n = 10$ with $\bar{x} = 85$, $s = 12$. Find the test statistic t.

a. 2.05

b. 3.05

c. **1.05 ✓**

d. 1.55

$SE = s/\sqrt{n} = 12/\sqrt{10} = 3.7947$. $t = \frac{85-81}{3.7947} = 1.05$.

35 [Intermediate] • Target: Chi-square test for homogeneity

In a hypothesis test, failing to reject H_0 when H_a is actually true. What is this called?

a. Confounding

b. **Type II error ✓**

c. Type I error

d. Sampling bias

This is a Type II error by definition.

© ClipSAT — clipsat.org | All Rights Reserved

36 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

In a hypothesis test, rejecting H_0 when H_0 is actually true. What is this called?

a. Confounding

b. **Type I error ✓**

c. Type II error

d. Sampling bias

This is a Type I error by definition.

37 **[Intermediate]** • **Target: Chi-square goodness-of-fit test**

A pharmaceutical company is investigating the efficacy of a new drug to reduce blood pressure. They want to design a randomized controlled experiment. Participants will be recruited from a specific patient population. To ensure generalizability and account for potential confounding factors, the researchers decide to stratify the participants by age group (Under 40, 40-60, Over 60) and by the severity of their hypertension (Mild, Moderate, Severe). If there are 150 participants in total and an equal number of participants are to be assigned to each treatment group (Drug A, Drug B, Placebo), and the number of participants in each stratum is proportional to the expected prevalence in the general population, how many participants would be allocated to the stratum 'Under 40' and 'Mild' hypertension if the proportion of individuals in this stratum is 0.08?

a. 10

b. **12 ✓**

c. 15

d. 8

The problem requires calculating the number of participants in a specific stratum within a larger experimental design. The total number of participants is 150. The participants are stratified by age group and hypertension severity. The proportion of individuals in the stratum 'Under 40' and 'Mild' hypertension is given as 0.08. To find the number of participants in this stratum, we multiply the total number of participants by the proportion of this stratum. Number of participants in 'Under 40' and 'Mild' stratum =.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

© ClipSAT — clipsat.org | All Rights Reserved

38 [Challenging] • Target: Chi-square test for independence • Cross-Lesson: 3.03 The Normal Distribution

What is the general structure of a 90% confidence interval?

a. estimate × (critical value) + (standard error)

b. (critical value) ± (standard error) × estimate

c. estimate – (critical value) ÷ (standard error)

d. **estimate ± (critical value) × (standard error) ✓**

A confidence interval always has the form: estimate ± margin of error, where margin of error = (critical value) × (standard error).

39 [Challenging] • Target: Interpreting chi-square results • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Increasing significance level α:

a. Decreases Type I error

b. **Increases power ✓**

c. Decreases power

d. Has no effect

Higher $\alpha \rightarrow$ more likely to reject $\rightarrow$ higher power (but more Type I errors).

40 [Challenging] • Target: Inference for the slope of a regression line • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A 95% confidence interval means:

a. 95% chance the true parameter is inside

b. **95% of all possible such intervals contain the true value ✓**

c. The sample mean is 95% accurate

d. p-value is 0.05

Frequentist interpretation: 95% of intervals capture μ.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Conditions for regression inference • Cross-Lesson: 1.01 Measures of Center & Spread

p-value is the probability of:

a. The null hypothesis being true

b. **Observing data at least as extreme, assuming H■ true ✓**

c. The alternative hypothesis being true

d. A Type II error

p-value is computed under the null.

42 [Challenging] • Target: Putting it together: a full inference scenario • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

The scatterplot below shows the number of hours a group of students studied for a test and their scores on that test. The line of best fit for the data is given by the equation $S = 5H + 40$, where S is the score on the test and H is the number of hours studied. If a student studied for x hours and achieved a score of 85, what is the value of x?

a. 10

b. 9

c. **8 ✓**

d. 7

The equation for the line of best fit is given as $S = 5H + 40$. We are told that a student studied for x hours and achieved a score of 85. We can substitute these values into the equation: $85 = 5x + 40$. To solve for x, first subtract 40 from both sides of the equation: $85 - 40 = 5x$ which simplifies to $45 = 5x$. Then, divide both sides by 5: $x = \frac{45}{5}$ which gives $x = 9$. Therefore, the student studied for 9 hours.

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Chi-square test for homogeneity • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

The scatterplot below shows the number of hours spent studying for a test versus the score achieved on that test for a group of students. The equation of the line of best fit is given by $S = 5H + 50$, where S is the score and H is the number of hours spent studying. If a student scored 80 on the test, approximately how many hours did they study?

a. 5.0 hours

b. 5.5 hours

c. **6.0 hours ✓**

d. 6.5 hours

The equation of the line of best fit is $S = 5H + 50$. We are given that a student scored $S = 80$. We need to find the number of hours H they studied. Substitute $S = 80$ into the equation: $80 = 5H + 50$. To solve for H, first subtract 50 from both sides of the equation: $80 - 50 = 5H$ $30 = 5H$. Now, divide both sides by 5: $H = \frac{30}{5}$. $H = 6$. Therefore, the student studied approximately 6.0 hours.

44 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 2.01 Sampling Methods & Bias

A line of best fit relating hours studied (x) to test score ($\hat{y}$) is $\hat{y} = 10x + 27$. Using this model, what is the predicted score for a student who studies 4 hours?

a. 63

b. **67 ✓**

c. 62

d. 76

Substitute $x = 4$ into the regression equation: $\hat{y} = 10(4) + 27 = 40 + 27 = 67$.

45 [Challenging] • Target: Chi-square goodness-of-fit test • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of ΔABC?

a. 12.5

b. **15 ✓**

c. 22.5

d. 30

To find the area of ΔABC, we first determine the coordinates of its three vertices: A, B, and C. 1. **Vertex C**: The function $f(x) = 9 - x^2$ is a downward-opening parabola with its maximum at $x = 0$. $f(0) = 9 - 0^2 = 9$, so the vertex is $C(0, 9)$. 2. **Intersection Points A and B**: Set $f(x) = g(x)$: $9 - x^2 = x + 3$ Rearrange into standard quadratic form: $x^2 + x - 6 = 0$ Factor the quadratic equation: $(x + 3)(x - 2) = 0$ Thus, the x-coordinates of the intersection points are.

© ClipSAT — clipsat.org | All Rights Reserved